

Final Project Proposal: Model Predictive Control for Energy-Aware Racing Line Optimization of Electric Vehicles

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1. Problem

Finding the optimal racing line on a sinuous closed track is a classical problem in motorsport engineering. On straights, every driver can floor it, so there is little room for differentiation. But in corners, performance differences emerge since the racing line governs the maximum speed a car can carry through a turn, and small cornering improvements translate into significant lap time gains (Xiong, 2010). Beyond the geometric question of which path to follow, the vehicle dynamics (tire grip, weight transfer, drift) make this a rich nonlinear optimization problem.

We add complexity by considering electric vehicles (EVs). Unlike conventional race cars, an EV must continuously manage its battery state of charge (*SOC*). Aggressive driving drains the battery faster through increased power demand, while conservative driving preserves energy but sacrifices lap time. This creates a genuine dual-objective problem as we want to go fast *and* be energy-efficient. In Formula E and the growing EV racing scene, this tradeoff directly determines race strategy.

The objective of this project is to develop an optimization-based controller that will compute racing trajectories for EVs on arbitrary closed circuits, jointly minimizing lap time and energy consumption. This problem naturally decomposes into two coupled optimization stages. The first is **path optimization** (*where* to drive), which will seek a racing line that exploits the full road width to reduce curvature through corners and allow higher cornering speeds. The second is **speed and control optimization** (*how* to drive along that path), which will determine the optimal speed profile and control inputs at every point while respecting tire grip, battery constraints, and the energy-time tradeoff. These two stages are coupled since the optimal speed depends on path curvature, and the best path depends on the vehicle's dynamic limits. Our approach will address both through a receding-horizon formulation.

This problem is well-suited to AA222 as it combines dynamic systems modeling, non-linear constrained optimization, and engineering design with competing objectives. It also connects to active research in eco-driving, autonomous racing, and predictive energy management (Sciarretta and Guzzella, 2007; Kochenderfer and Wheeler, 2019).

2. Proposed Approach

We plan to formulate the problem as a finite-horizon *Model Predictive Control* (MPC) problem. At each timestep, the controller will solve an optimization over a receding horizon, apply the first control action, then re-solve using updated state information.

Vehicle Model

We will use a planar bicycle model capturing both longitudinal and lateral dynamics while remaining tractable. The state is $\mathbf{z} = [x, y, \psi, v_x, v_y, \omega, SOC]^\top$ (position, heading, velocities, yaw rate, battery charge) and the control inputs are $\mathbf{u} = [\delta, F_{\text{drive}}]^\top$, that is steering angle and net longitudinal force (positive for traction, negative for regenerative braking).

The longitudinal dynamics follow $m\dot{v}_x = F_{\text{drive}} - \frac{1}{2}\rho C_d A v_x^2 - C_r mg + m v_y \omega$, where $\rho C_d A$ captures aerodynamic drag and C_r is rolling resistance. Lateral and yaw dynamics will be governed by tire forces through a simplified Pacejka-like model, with the friction circle constraint $F_x^2 + F_y^2 \leq (\mu mg)^2$ ensuring the car does not exceed grip. The battery state evolves as $\dot{SOC} = -P_{\text{elec}}/(Q_{\text{batt}} \cdot V_{\text{nom}})$, where $P_{\text{elec}} = F_{\text{drive}} \cdot v_x/\eta$, with motor efficiency η (accounting for regen when $F_{\text{drive}} < 0$), battery capacity Q_{batt} , and nominal voltage V_{nom} .

Optimization Formulation

For the **path optimization stage**, we will parameterize the racing line by lateral offsets $\alpha = [\alpha_1, \dots, \alpha_N]$ from the track centerline at N equally spaced stations along the circuit, so that each point on the path is $\mathbf{p}_i = \mathbf{c}_i + \alpha_i \hat{\mathbf{n}}_i$ where $\mathbf{c}_i$ is the centerline position and $\hat{\mathbf{n}}_i$ the outward normal. The constraint $|\alpha_i| \leq w_i/2$ ensures the path stays within the track boundaries of half-width w_i . We will then search over α to minimize the overall cost J defined below.

For the **speed and control stage**, given a candidate path, we will minimize the weighted cost

$$J(\alpha, \mathbf{u}) = \underbrace{w_t \int_0^T 1 dt}_{\text{time}} + \underbrace{w_e \cdot \Delta SOC}_{\text{energy}} + \underbrace{w_u \int_0^T \|\dot{\mathbf{u}}\|^2 dt}_{\text{smoothness}}$$

subject to speed bounds $\|\mathbf{v}\| \leq v_{\text{max}}$, actuation limits $|F_{\text{drive}}| \leq F_{\text{max}}$ and $|\delta| \leq \delta_{\text{max}}$, battery limits $SOC_{\text{min}} \leq SOC \leq 1$, tire grip $F_x^2 + F_y^2 \leq (\mu mg)^2$, and track boundary constraints $|\alpha_i| \leq w_i/2$. Since the two stages are coupled, we will iterate between them or solve jointly. Varying w_t and w_e will let us explore the Pareto front between lap time and energy consumption.

Experimental Setup

We plan to test the controller on several simulated closed circuits of varying complexity, from simple ovals to tracks with tight hairpins and chicanes, as shown in fig. 1.

Baselines. We will benchmark our MPC controller against four baselines, namely constant-speed center-line following, aggressive minimum-time driving that ignores energy, minimum-curvature path planning (Heilmeier et al., 2020), and a heuristic energy-saving strategy.

Metrics. We will evaluate each approach on lap time, energy consumed ΔSOC , ride smoothness $\int \|\dot{\mathbf{u}}\|^2 dt$, and solver runtime. We also plan to construct Pareto fronts between lap time and energy use to characterize the tradeoff.

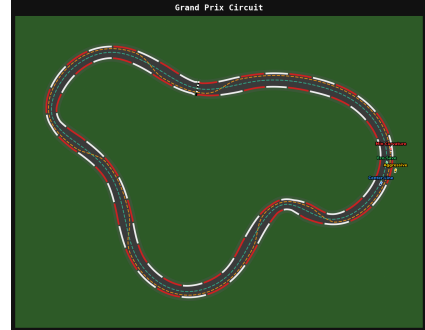


Figure 1: Simulated circuit with track boundaries and center-line.

References

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